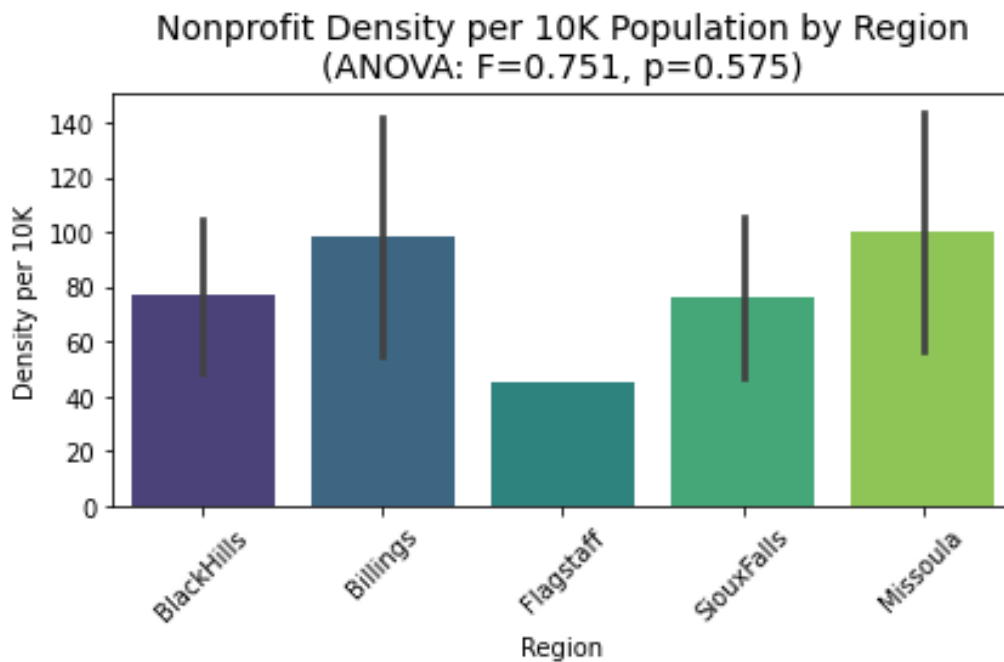


Q1: Is there a significant difference in the density of nonprofit organizations between Black Hills and benchmark regions?

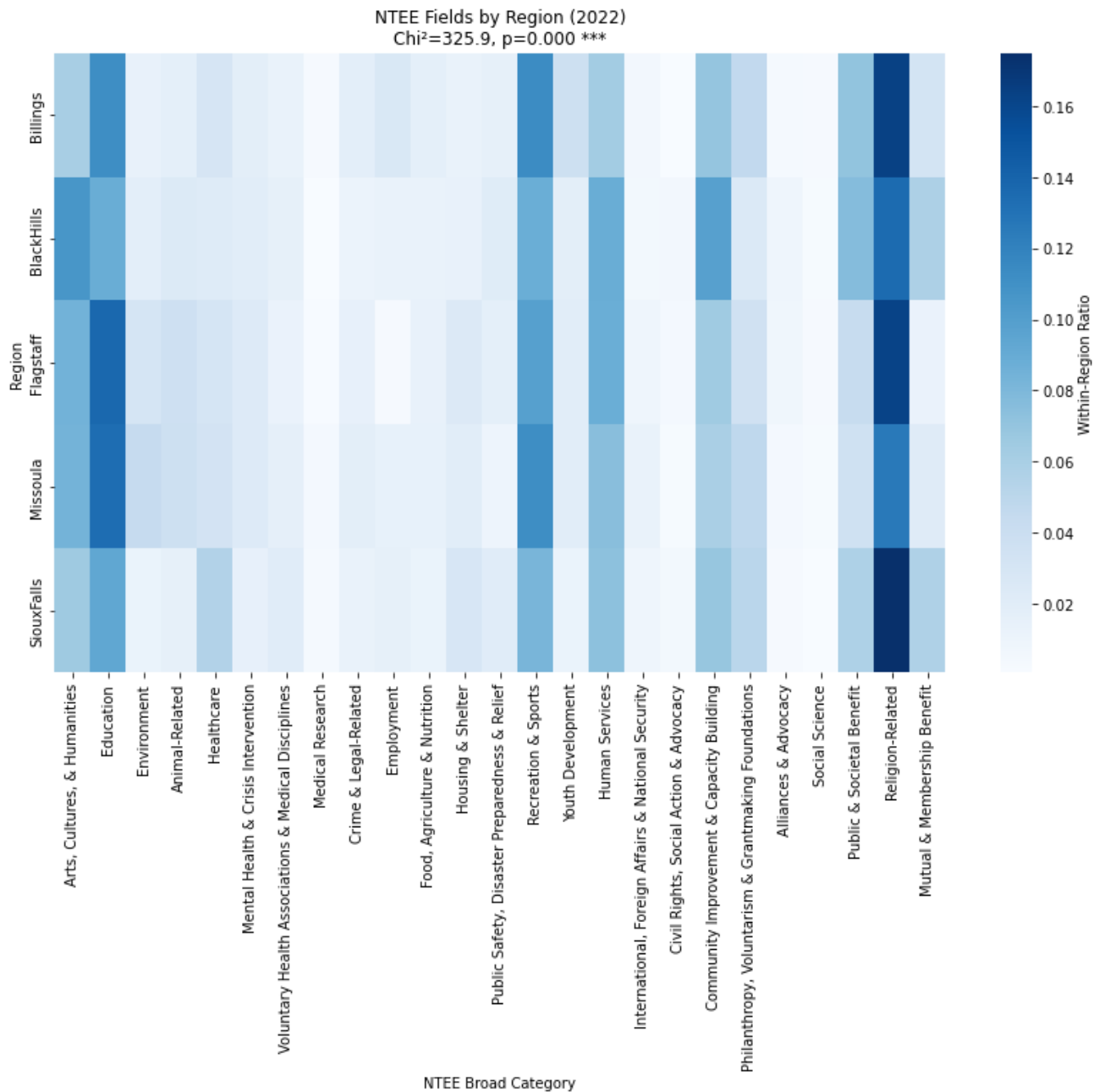
- Density per 10K: number of nonprofit organizations per 10K population in each county
- Each county is mapped to Black Hills or benchmark regions



- The height of the bars indicates the mean nonprofit density per 10K population for each region
- The error bars indicate 95% confidence intervals
- There is no statistically significant difference of nonprofit density across the five regions

Q6: Are there any nonprofit fields under/over-represented in each region?

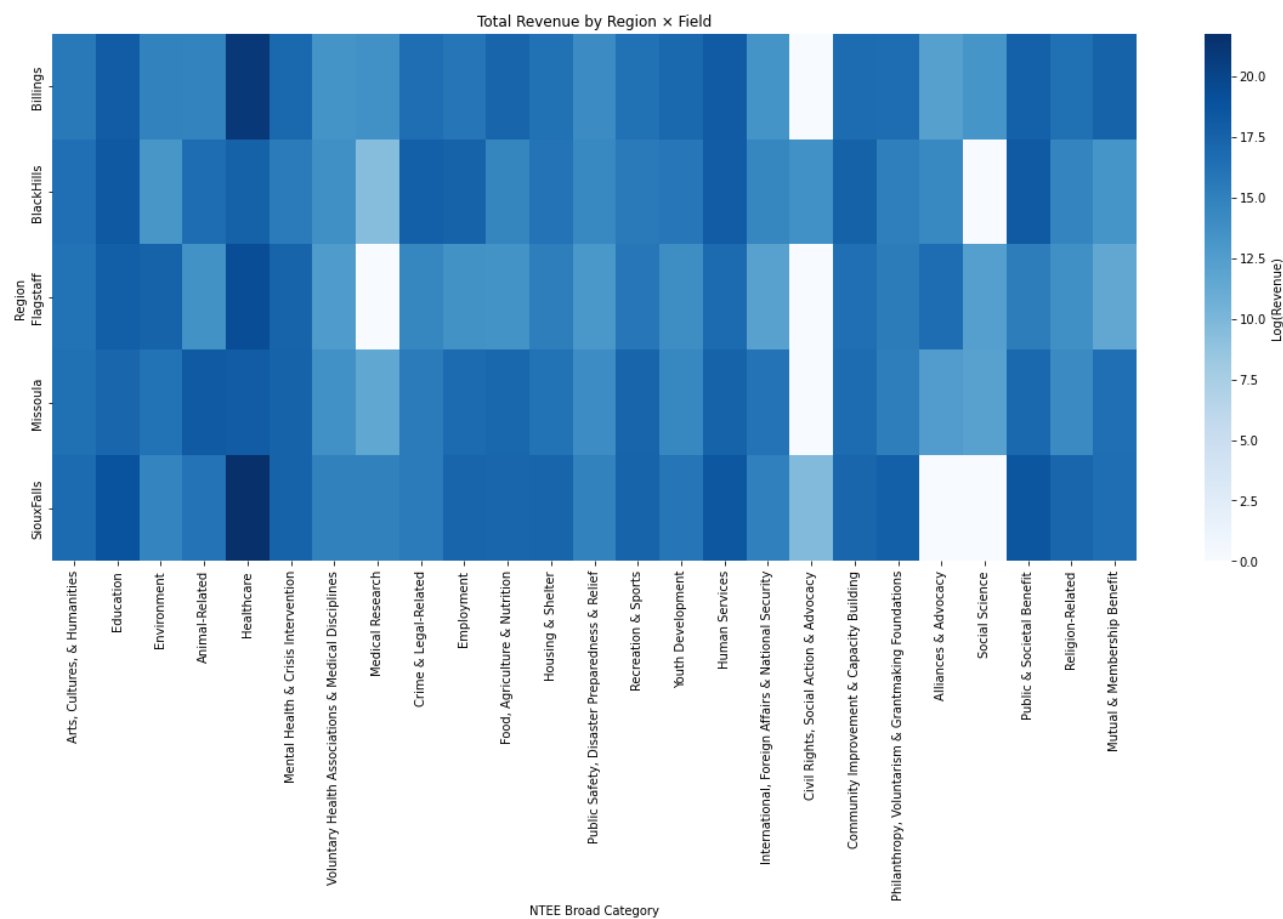
- Using NTEE classification of business fields
- Within-Region Ratio = (nonprofit counts in the field)/(total nonprofit counts for the region)



- Unidentified nonprofits were excluded (26, 24, 11, 20, 56 respectively)
- Unknown nonprofits were also excluded (4, 0, 0, 1, 6 respectively)
- Billings, Flagstaff, and Sioux Falls have proportionally more religion-related nonprofits (more than 16% of nonprofits in each region vs. 12-13% for other regions)
- Sioux Falls has more healthcare nonprofits than the other four regions
- Revenue can be a very high number, and the visualization is dominated by the few highest revenues:

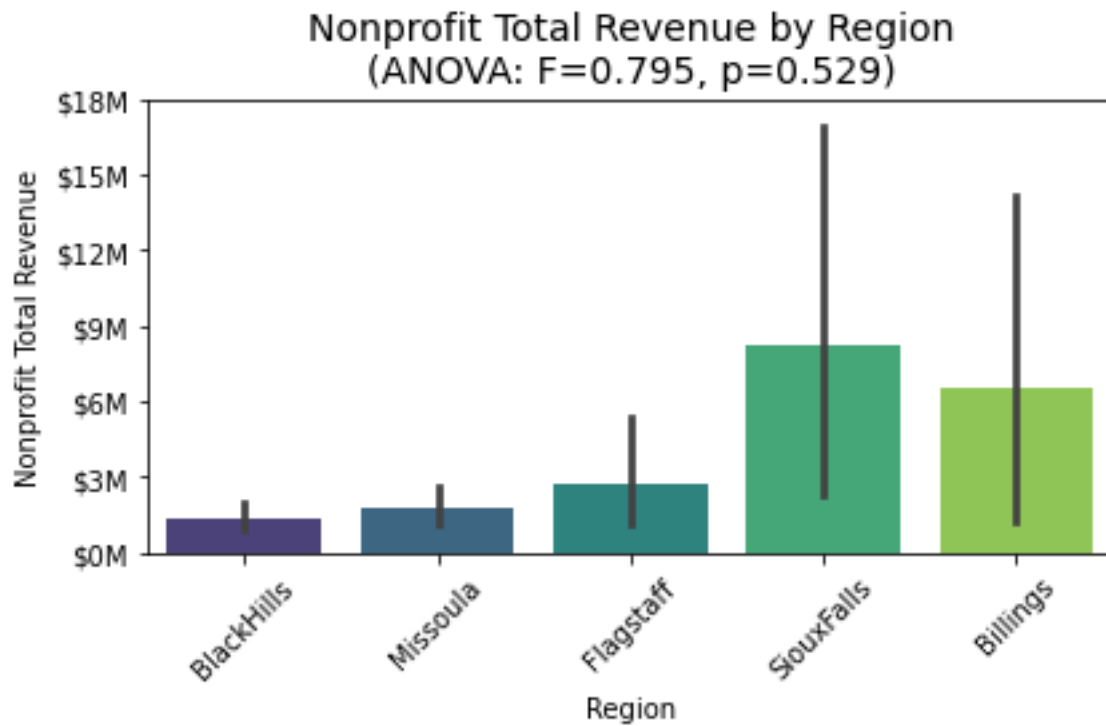


- Taking the logarithm of (revenue +1) reduces the difference between the very high values and smaller values:

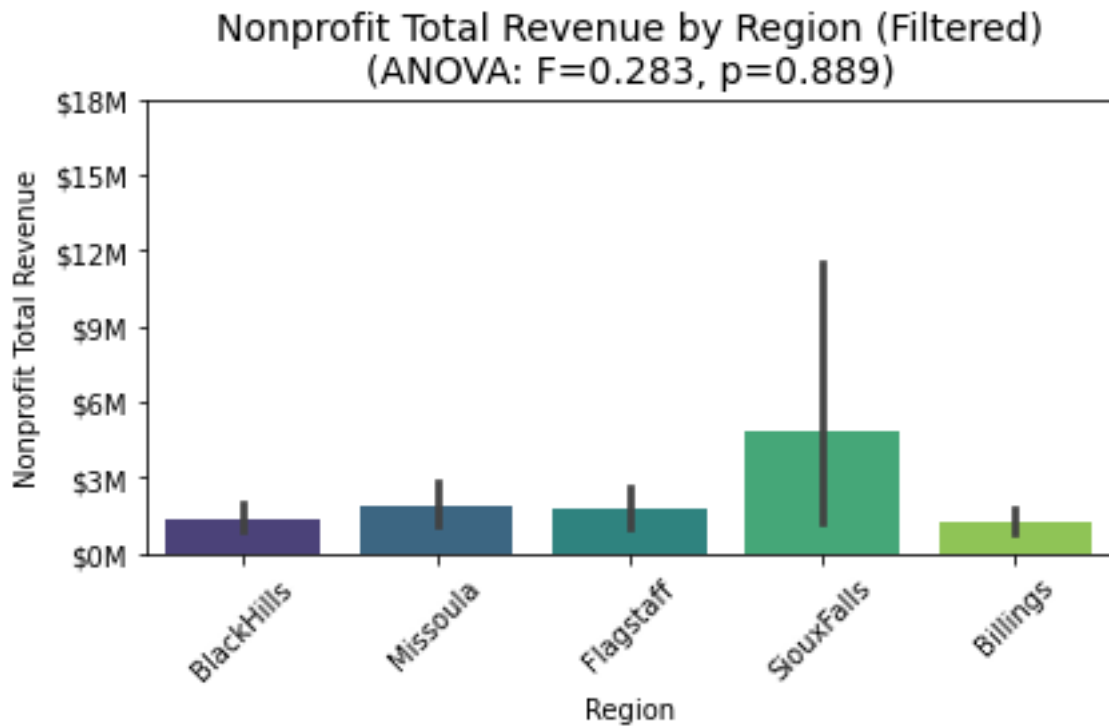


- Billings and Sioux Falls get higher total revenue in the healthcare field than the other three regions

Q3: Is there a significant difference in the total revenue of nonprofit organizations between Black Hills and benchmark regions?

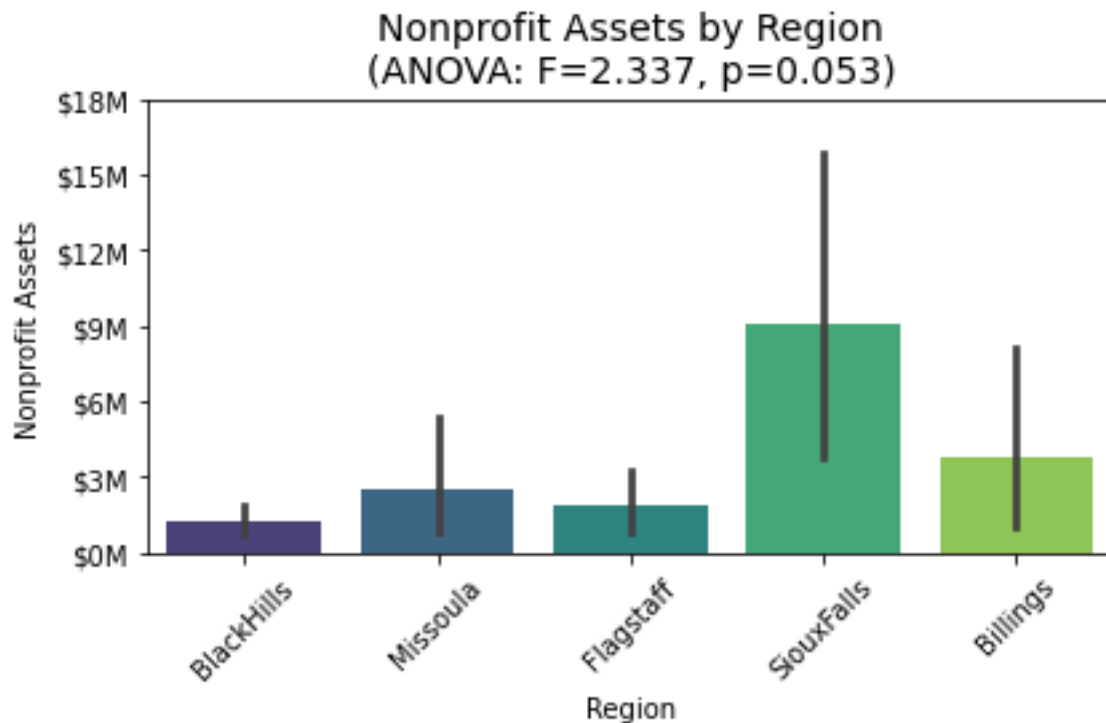


- There is no statistical significance across the five regions, but Black Hills has the lowest average total revenue. It is important to note that there is practical significance, especially with Sioux Falls and Billings.

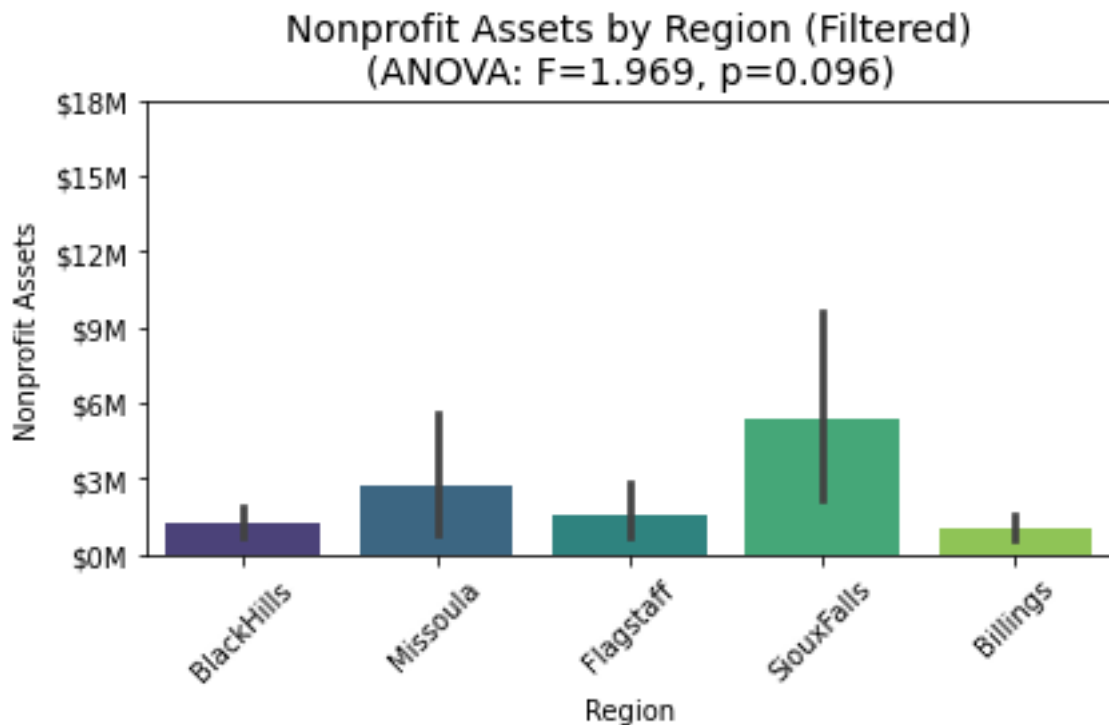


- Filtered data excludes hospitals, universities, and political (501c4) organizations. The average total revenues for Sioux Falls and Billings decrease significantly and become closer to the average total revenue of Black Hills. Based on the results from Q6, this might be due to its larger presence in the hospital industry.

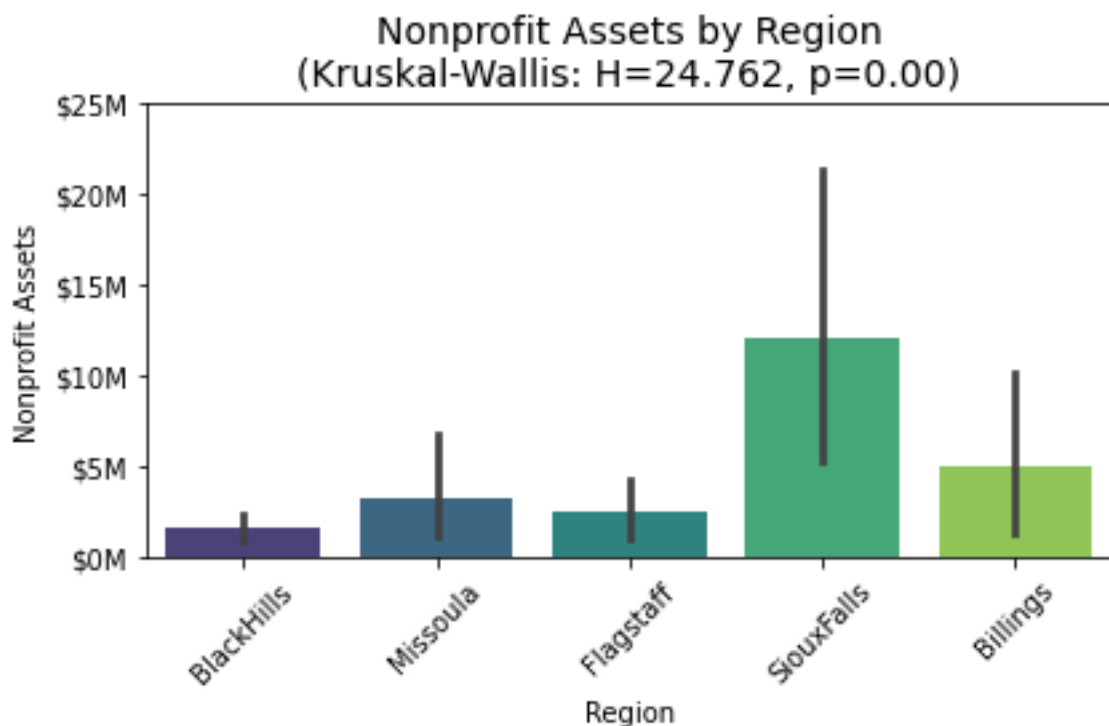
Q4: Is there a significant difference in the assets of nonprofit organizations between Black Hills and benchmark regions?

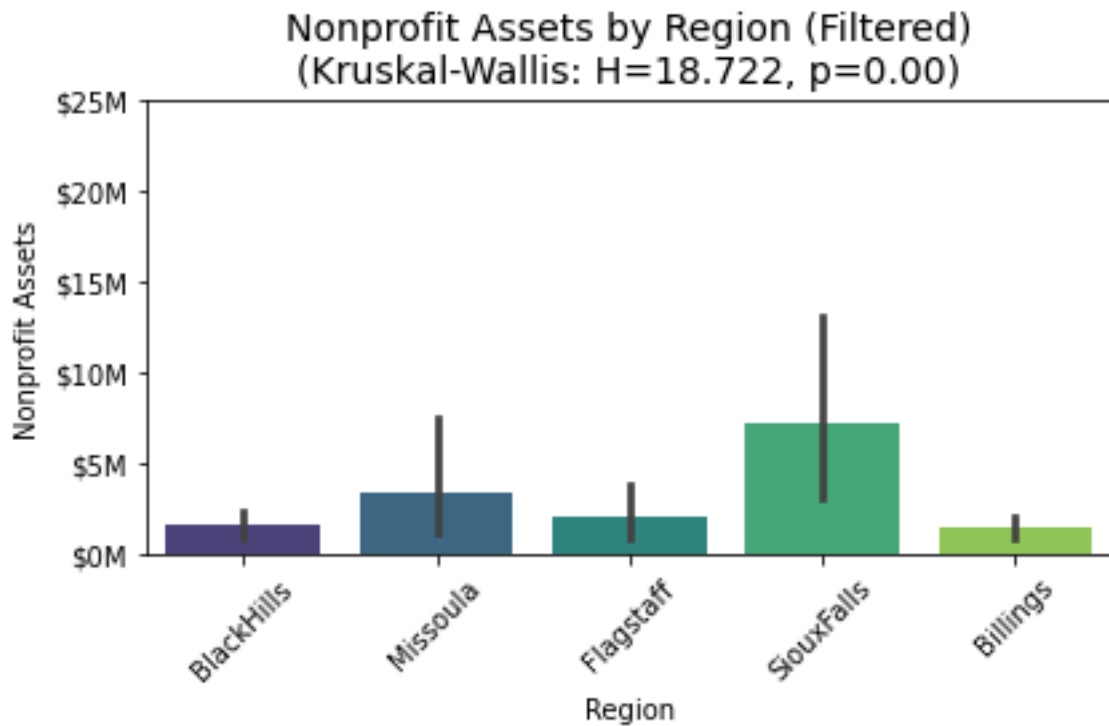


- There is no statistically significant difference of nonprofit assets across the five regions if we set $\alpha=0.05$. However, it is important to note the difference between statistical and practical significance. Sioux Falls has, on average, more assets than other regions.

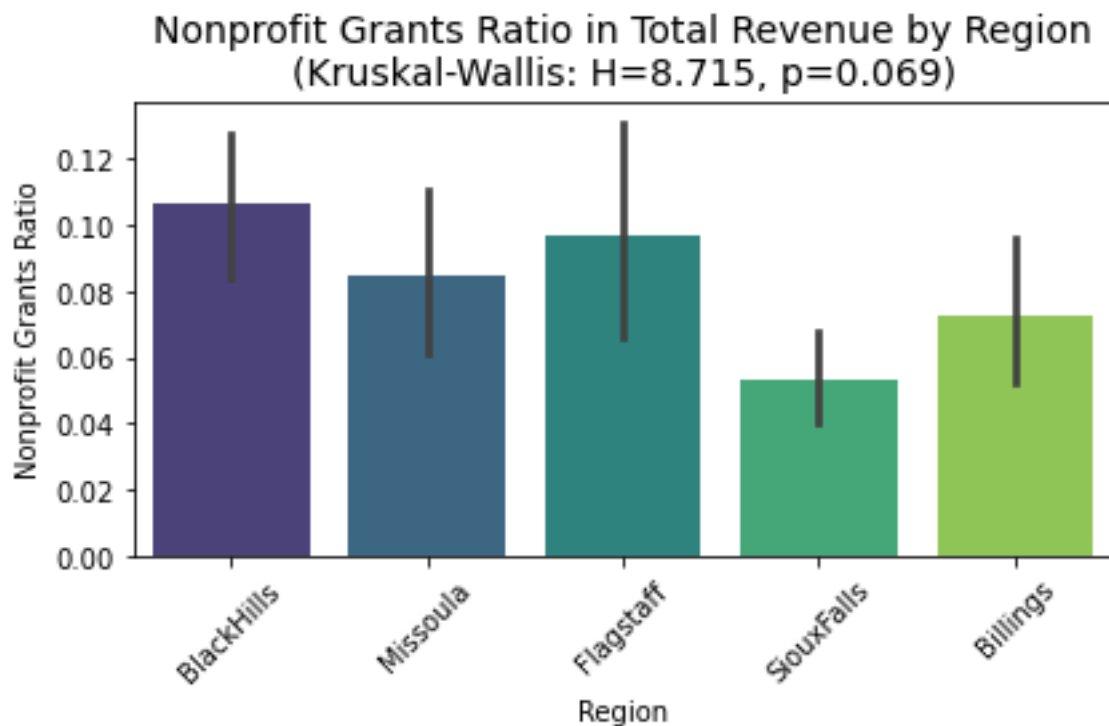


- Filtered data excludes hospitals, universities, and political (501c4) organizations. The significance is smaller, but Sioux Falls still has the highest average assets. The average assets for Black Hills do not decrease as much as Sioux Falls and Billings. Based on the results from Q6, this might be due to the fact that Sioux Falls and Billings have a larger hospital industry.

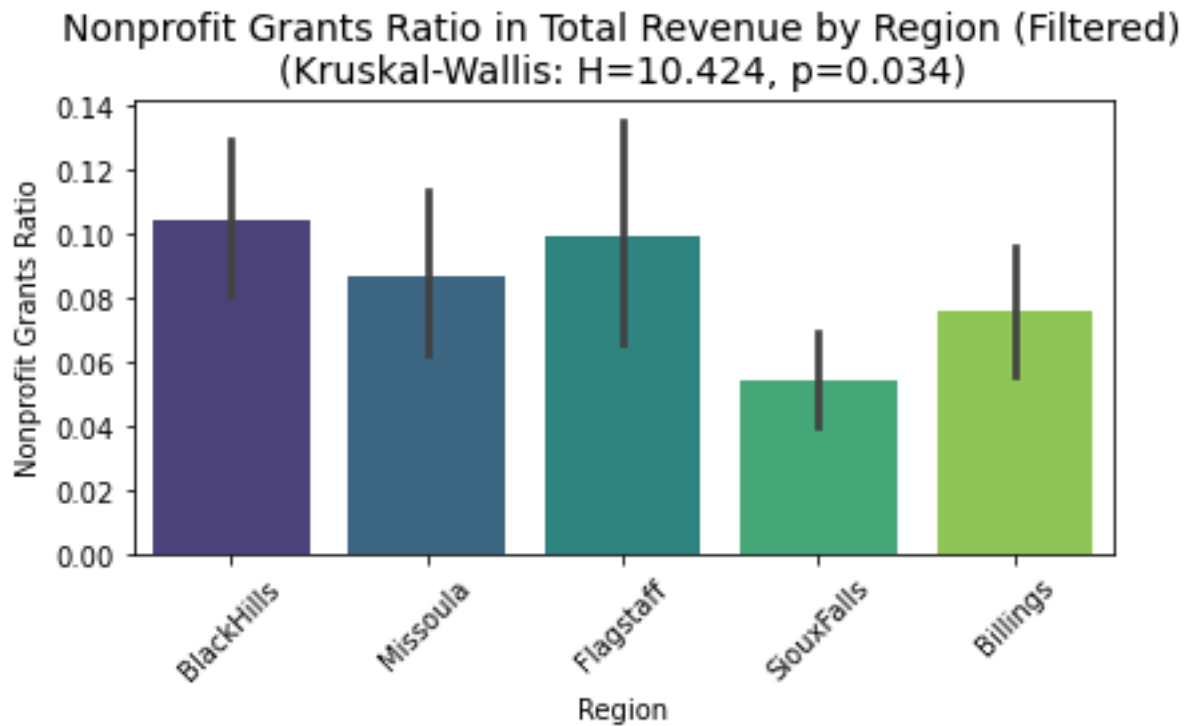




Q5: Is there a significant difference in nonprofits' dependency on grants vs. other revenue sources between Black Hills and benchmark regions?

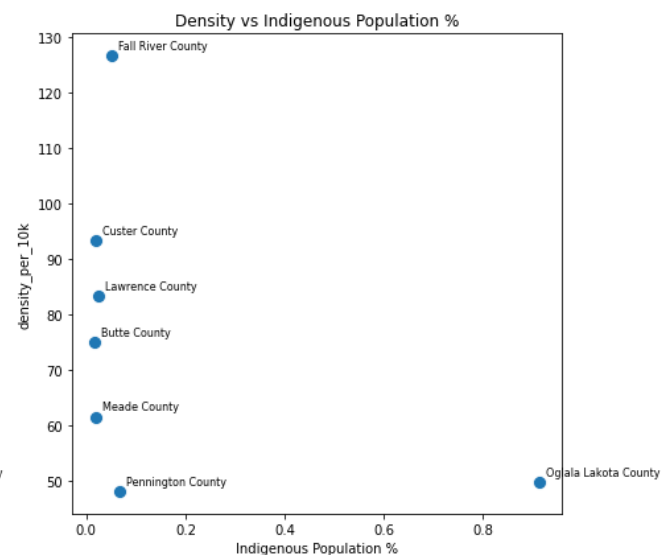
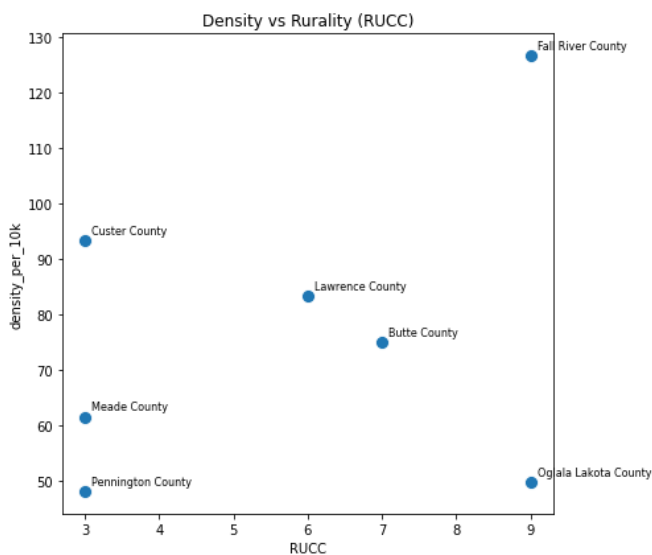


- Grant ratio = (Grant amount) / (Total revenue)
- Kruskal-Wallis as an alternative to ANOVA, data does not meet the normality assumption
- Black Hills has proportionally more grants in their total revenue than Sioux Falls.
- Shorter error bars indicate less variability within regions compared to other metrics.



Q2: Do counties with higher indigenous population and rurality have fewer nonprofit organizations in the Black Hills region?

- Higher RUCC = more rural



- Oglala Lakota County has a low nonprofit density compared to other BH counties. Excluding the Oglala Lakota County, there seems to be no association between Native American population and nonprofit density.
- Fall River County has the highest nonprofit density despite having the same rurality score as Oglala Lakota County. Excluding these two counties would show no association between rurality and nonprofit density.

